

4.5.1. Message Length (16/32 bits)

An optional unsigned integer value, in little-endian byte order, specifying the overall length of the message in bytes, not including the size of this field itself. The size of this field, when present, SHALL depend on the transport protocol being used to transfer the message. For example, for TCP, it SHALL be set to 4 bytes to allow for large payloads, whereas for PAFTP and BTP it SHALL be set to 2 bytes.

Protocol	Size	Location Description in Stream
TCP	4 bytes	Field present before the Message Header of each Matter message.
BTP	2 bytes	Field present before the segment payload in the beginning BTP Packet PDU for each Matter message/BTP SDU.
PAFTP	2 bytes	Field present before the segment payload in the beginning PAFTP Packet PDU for each Matter message/PAFTP SDU.
MRP	N/A	Field not present.
NTL	N/A	Field not present.

4.6. Message Counters

All messages contain a 32-bit message counter assigned by the sender of the message. Message counters are assigned sequentially, by monotonically increasing the counter value maintained by the sender of the message. Message counters serve several purposes:

- **Duplicate Message Detection** – Receiving systems use message counters to detect messages that have been retransmitted by the sender, e.g. in response to packet loss in the network.
- **Message Acknowledgement** – In the Message Reliability Protocol (MRP), message counters provide a way for receivers to identify messages for the purpose of acknowledging their receipt.
- **Encryption Nonces** – When encrypted messages are sent, message counters provide an encryption nonce that ensures each message is encrypted in a unique manner.
- **Replay Prevention** – Related to encryption, message counters also provide a means for detecting and preventing the replay of encrypted messages.

4.6.1. Message Counter Types

All Nodes implement three global 32-bit counters to generate message counters for certain types of messages:

- *Global Unencrypted Message Counter*
- *Global Group Encrypted Data Message Counter*
- *Global Group Encrypted Control Message Counter*

Additionally, Nodes implement a separate 32-bit counter for each session as part of secure session state:

- *Secure Session Message Counter*